

The New York Public Library

Jail & Prison Services
40 West 20th Street
New York, NY 10011

DATE, 2024

FirstName LastName #12345678
Whatever Correctional Facility
P.O. Box 12345
New York, NY 12345

Dear (Insert Name),

Thank you for your letter asking for ideas to scale up your small bottle recycling company in New York City. You included questions about promising technologies or equipment, how AI is applied to bottle recycling, and possible add-ons to your existing company. In your next letter, it would be helpful to include a quick description of the machinery, workspace, and job roles or departments you already have. If any information was redundant or not in-depth enough, please write back to us with more questions!

Section I. General Tips and Ideas for Growing Your Business.....2

- A. Get more funding
- B. Expand to new locations
- C. Merge and acquire businesses

Section II. New and Promising Technologies for Bottle Recycling5

- A. Before buying new equipment, here are some things to consider
- B. Innovations in sorting technology
- C. Efficient and advanced washing lines
- D. Artificial intelligence integration in recycling
- E. Manufacturers and suppliers to take note of
- F. Chemical recycling
- G. Biorecycling

Section III. Add-Ons14

- A. Recycling more than just plastic bottles
- B. Marketing, social media, customer database management

Section IV. Further Resources16

- A. Books
- B. Contacts

Please let us know if we can be of any assistance in the future. And, we've changed our address, please direct all future letters to Jail & Prison Services, 40 West 20th Street, New York, NY 10011.

Sincerely,
Jace

Section I. General Tips and Ideas for Growing Your Business

Sourced from <https://www.sba.gov/business-guide>

A. Get more funding

If your business is up and running but needs more capital, you can rely on familiar options. However, funding an existing business still requires slightly different preparation.

i. Prepare to request more funding

Anyone who gives you funds wants to feel confident that their investment will pay off. Prepare a **business case** and **financial statements** to convince lenders, crowd funders, or investors to fund your small business.

Make your business case: You'll need to make a solid business case for more funding. Produce a short statement with the total requested amount and specific reasons for it. Maybe your business is cyclical — like construction or education — and could use funding to get through expected slow periods. Or maybe it needs capital to **invest in new machinery** or launch a product line. Whatever the reason, update your business plan to include this stage of funding.

A business case should give assurances that new funds won't be mismanaged. Include descriptions of your management team to **highlight their skills and expertise**.

Prepare financial statements: Display that your business is doing well with **financial history statements**. Show how your business has grown by reporting revenue, expenses, and profit over time. If you don't have a history of positive growth, explain why more funding will allow you turn it around.

- Prove you're financially responsible with a **business credit report**. Review your business credit file to make sure it's accurate before sharing it.
- Determine how much your company is worth today by performing a **business valuation**. This is the same process you'd go through if you were planning to sell your business. Valuation methods vary, but you can do a self-evaluation or seek out a qualified business appraiser.
- Show how your business will grow in the future with a **forecast**. Your business forecast can be based on intuitive judgment, quantitative analysis, or both. Show your projected revenue and expenses, and clearly explain how you arrived at those estimations.

Connect with a local SBA resource center: Meet with local experts, counselors, and business mentors at a local SBA (Small Business Administration) resource center (<https://www.sba.gov/local-assistance>) if you need help preparing your business to get more funding.

ii. Choose your funding source

Get loans, credit, or crowdfunding: Additional funding options for existing businesses are similar to funding options for a new business. You'll have the same general set of options, which include small **business loans, credit cards, and crowdfunding**.

Existing businesses have the advantage of an established financial history with credit reports, business bank accounts, and internal financial reports. Lenders, investors, and even crowdfunders can use that information when they decide whether to fund your business.

On **September 16, 2024**, the **EPA** announced **three funding opportunities** for recycling and wasted food prevention:

- Recycling Education and Outreach Grant Program
- Solid Waste Infrastructure for Recycling Grants for Communities
- Solid Waste Infrastructure for Recycling Grants for Tribes and Intertribal Consortia

Use Lender Match to find lenders who offer SBA-guaranteed loans: If you have trouble getting a traditional business loan, look into SBA-guaranteed loans. When a bank thinks your business is too risky to lend money, SBA may guarantee your loan — that way the bank has less risk and could be more willing. Use Lender Match (<https://www.sba.gov/funding-programs/loans/lender-match-connects-you-lenders>) to find lenders who offer SBA-guaranteed loans.

B. Expand to new locations

Once you're ready to expand, update your marketing plan and confirm that your business is financially prepared. Then, make sure to comply with all laws, rules, and regulations in the new business locations.

i. Prepare for a new market

First, update your marketing plan with your new location in mind. Think about your **target customer, sales plan, and competitive advantage**. Add up any additional marketing and sales costs. Make sure your updated marketing plan is just as thorough as your initial plan.

Compare your business to the competition, learn about the local market, and get a sense of the advertising market.

Next, review your business finances. Build a **forecast** that projects estimated costs and estimated revenue

Preparation

1 Update marketing plan.



2 Check finances.



Legal

4 Get foreign qualification.



3 Get licenses and permits.



5 Pay new taxes.



for your new location. Take a close look at your **balance sheet** to make sure you can cover the costs of expanding. If you don't have enough capital, you can try to get more funding.

ii. Legal steps to expand your business

Expanding your business to a new state, county, or city isn't very different from opening a new business there. You'll want to make sure you **register** your business with the right agencies and pay the appropriate taxes.

License, permit, and zoning rules: These rules vary across states and localities. Getting licenses and permits in new locations is similar to getting them in your home state.

If you already have a permit or license from a federal agency, check with the issuing agency to confirm you can legally operate in a new state. Also, see whether your new state, county, and city governments require a new license or permit. Start by visiting your state's website.

Foreign qualification: If you plan to expand your business to a new state, you might need to file for foreign qualification in that state. This process notifies the new state that your business is active there.

To foreign qualify, file a **Certificate of Authority**. Many states also require a **Certificate of Good Standing** from your state of formation. Each state charges a filing fee, but the amount varies by location and business structure. Check with state offices to find out foreign qualification requirements and fees.

Pay taxes in new states and localities: If you do business in a new state as a foreign qualified business, you'll typically need to **pay taxes** and **annual report fees** in the new state as well as your home state. The process for foreign qualified businesses to pay taxes is similar to any other business that needs to pay taxes in the state.

Keep in mind that not every state and locality has a sales tax. In addition, most states have **tax exemptions** on certain items, such as food or clothing. If you charge sales tax, you need to be familiar with **applicable rates**.

C. Merge and acquire businesses

You can grow your business by buying or merging with a smaller business. The process is similar to starting a new business, but you need to take extra steps to protect your existing business.

i. Differences between mergers and acquisitions

Mergers combine two separate businesses into a **single new legal entity**. True mergers are uncommon because it's rare for two equal companies to mutually benefit from combining resources and staff, including their CEOs. Unlike mergers, **acquisitions** do not result in the formation of a new company. Instead, the purchased company gets **fully absorbed by the acquiring company**. Sometimes this means the acquired company gets liquidated. Acquiring a business is similar to buying an existing business or franchise.

Calculate how much the other business is worth: Conduct a **business valuation** to determine the value of the other business before you agree to a sale. This is essentially the same process

you'd go through to figure out how much your own business is worth before closing or selling your business.

There are several ways to value a business, so do extensive research on methods if you choose to do it on your own. You might want to **hire a qualified business appraiser**. Once you know how much the other business is worth, you'll know whether you can afford it outright or if you need to get more funding.

Make a merger or acquisition agreement: You must prepare a **sales agreement** to move forward with the sale or merger. This document allows for the purchase of assets or stock of a corporation. An **attorney** should review it to make sure it's accurate and comprehensive.

List all inventory in the sale along with names of the businesses and owners. Fill in the relevant background details. Determine how the business will be run prior to close and the level of access each company will have to financial information. Note all **adjustments, broker fees, and any other aspects relevant to the terms of agreement**. Don't leave out any assets and liabilities, or this can create problems even after the sale has been finalized.

Transfer business ownership: The terms of your agreement will dictate which steps you must take to transfer ownership, and what that ownership will look like. It's widely recommended to have an attorney help with this step.

After you've completed the acquisition or merger, you'll need to **register** these changes with the state, depending on state law and business structure. If the merger requires you to dissolve your original company and create a new one, you might also need to open new business bank accounts, get new state and federal tax IDs, re-apply for licenses and permits, and take steps to legally close your old business.

Section II. New and Promising Technologies for Bottle Recycling

A. Before buying new equipment, here are some things to consider:

The type of plastic you recycle: You need to consider what type of plastic you want to recycle.

Quantity: You will need to consider how much material you want to recycle. This will help you decide on the size and capacity of the machine you need.

Space: You need to consider where you will be placing your equipment. You need to make sure you have enough space for the machine and any products you want to recycle.

Budget: When buying plastic recycling equipment you need to consider your budget. Make sure you are getting the best value for your money.

Equipments Supplier: A good supplier will provide you with quality equipment at a reasonable price and support you when you need it. The right supplier will help you get the most out of your recycling business. For example by offering more **professional advice** or providing good **after-sales service** down the road.

i. Automated vs. Manual Intervention

PET bottle recycling lines can be categorized based on their level of automation. Some systems operate fully automatically, minimizing the need for manual intervention throughout the process. On the other hand, certain stages in other recycling lines may require manual sorting or quality control checks.

Fully automated PET bottle recycling lines offer numerous benefits:

- **Efficiency:** Automation ensures consistent and continuous operation without interruptions caused by human factors, making it ideal for processing pet flakes.
- **Higher throughput:** Automated systems can handle larger volumes of plastic waste at a faster pace.
- **Reduced labor costs:** With fewer manual tasks involved, businesses can optimize their workforce while maintaining productivity.

Hybrid machines take recycling to a whole new level by combining multiple functions for comprehensive waste management. These versatile machines not only crush, compact, and sort plastic bottles but also incorporate additional features like washing and drying capabilities. They provide an **all-in-one solution** for recycling plastic bottles efficiently. With hybrid machines, you can witness the transformation of used bottles into clean, compacted bundles ready for further processing or reuse.

By embracing the concept of hybridity, these machines contribute significantly to reducing the environmental impact of plastic waste. Instead of relying on separate devices for different stages of recycling, hybrid machines consolidate everything into a single unit, maximizing efficiency while **minimizing resources and space requirements**.

However, it's worth noting that **certain stages within the recycling process may still require manual inspection** or intervention even in fully automated systems. This ensures quality control and helps identify any contaminants or non-recyclable items.

ii. Smaller scale additions to your current recycling line

Adding a metal detector to the conveyer

belt: Metal detectors can replace the manual detection of each plastic, when a hard metal is detected, the metal detector will **issue an alarm** and let the conveyor belt **stop working temporarily** to protect the plastic recycling crusher machine does not return because of the wear and tear of the blades caused by crushing metal, which plays a good role in **protecting** the back of the equipment.



If expading plastic recycling beyond bottles: Larger plastic waste can be pre-processed by adding a shredder at the front of the plastic recycling line.



iii. Ordering international products

HS Code: HS codes, or Harmonized System codes, are a standardized system used globally to **classify and categorize products for international trade**. These codes consist of six digits, with each digit representing a specific level of classification. You can easily identify the different categories of plastic recycling machines by looking at their respective HS codes. These codes are a crucial aspect of international trade regulations, allowing for easy classification and identification of products.

Code	Description
84	Machinery and Mechanical Appliances
84.74	Industrial machinery for plastics processing
84.74.20	Plastic recycling machines

Tariffs: To navigate the complexities of tariff classification for plastic recycling machines, you'll need to be aware of specific **criteria and regulations**. Improper classification can have serious implications such as higher import duties or delays in customs clearance.

Here are three key challenges in tariff classification:

- **Ambiguous descriptions:** Some HS codes may not provide clear guidance on classifying plastic recycling machines, making it difficult to determine the correct code.
- **Constantly evolving technology:** The rapid advancements in recycling technology often result in new equipment with unique features and functions. This creates a challenge in finding the appropriate HS code that accurately represents these machines.
- **Varying interpretations:** Different countries may have their own interpretation of HS codes, leading to discrepancies and potential disputes during customs inspections.

B. Innovations in Sorting Technology

i. Optical Sorting Systems

Optical sorting systems are advancements that have significantly improved the **accuracy** of material separation in recycling. These systems use **sensors and cameras** to identify different types of materials on waste logistics conveyor belts. By analyzing the heat map of an item, waste logistics systems can efficiently **separate various materials** such as plastics, glass, and metals.

The importance of optical sorting and waste logistics is evident in its ability to increase the **efficiency and effectiveness** of recycling facilities. For example, it enables the identification and separation of **different colors** of plastic bottles with high precision. This process ensures that each type of material is sent for appropriate recycling processes, **reducing contamination** and increasing the overall **quality** of recycled materials.

ii. Robotics in Recycling Facilities

The integration of robotics has revolutionized the sorting process in recycling facilities. With **robotic arms equipped with advanced sensors**, these machines can swiftly pick out specific items from conveyor belts based on **predefined criteria**. This



automation not only speeds up the sorting process but also reduces errors while handling a wide variety of recyclable items.

Robotic technology plays a top role in scaling up recycling operations by efficiently managing **large volumes** of incoming materials. The ability to handle complex tasks like separating mixed paper from cardboard or removing contaminants from recyclables showcases how robotics contribute to streamlining recycling operations at a significant scale.

Fanuc, an international robotics company with global headquarters based in Japan, partnered with **Recycleye**, a tech start-up company based in London, to develop Recycleye Robotics. The machine can identify, sort and pick materials at a rate of 55 picks per minute compared with 40 to 45 picks by humans. The machine has several applications, including negative sorting. It can sort materials like polyethylene terephthalate (PET), high-density polyethylene (HDPE), polyethylene/polypropylene (PE/PP), aluminum and paper.

iii. Near-Infrared Technology

Near-infrared (NIR) technology has enhanced the identification process for recyclable materials by providing detailed information about their composition. By analyzing reflected light spectra, NIR technology **accurately identifies different types of polymers within plastics**, enabling precise sorting based on polymer composition.

The advances in NIR technology highlight its significance in **improving material purity** during recycling processes. It ensures that only high-quality recyclable materials are selected for further processing while effectively removing non-recyclable contaminants.

C. Efficient and advanced washing lines

High-speed friction washer: This powerful machine uses friction to remove contaminants from the plastic flakes. As the flakes pass through the washer, they undergo intense agitation, scrubbing away any remaining dirt or residue. The friction washer ensures that even **stubborn impurities are effectively dislodged**, leaving behind clean and pristine plastic flakes.

Efficient water reuse with filtration systems: In addition to their cleaning capabilities, some pet bottle washing lines feature water filtration systems that allow for efficient water reuse during the cleaning process. These systems help **conserve water resources** while maintaining optimal performance.

The filtration system works by **removing particles and impurities from the used water** before it is recycled back into the washing line. This not only reduces water consumption but also ensures that the cleaning process remains effective throughout operation. By reusing filtered water, these washing lines minimize environmental impact without compromising on cleanliness.

D. Artificial Intelligence Integration in Recycling

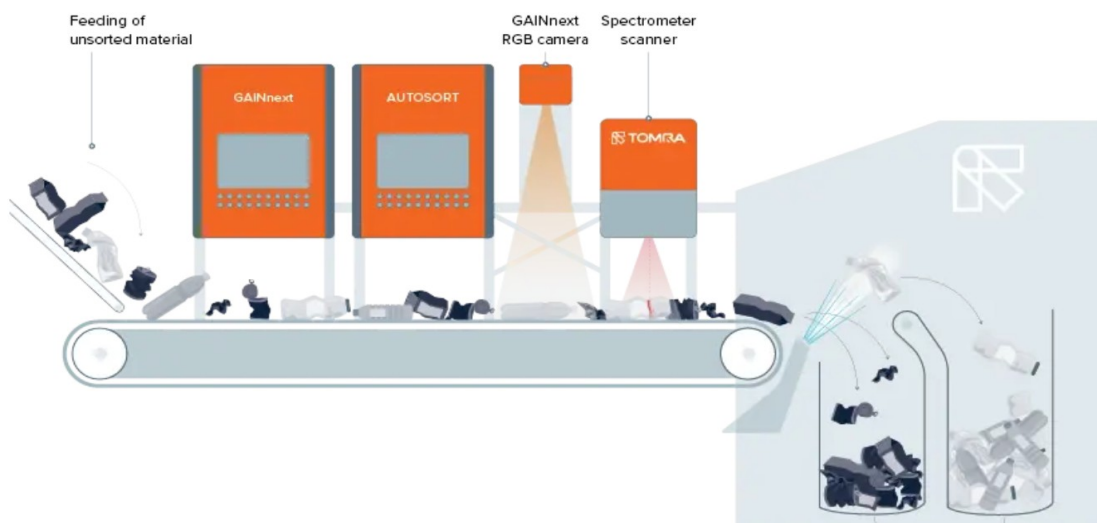
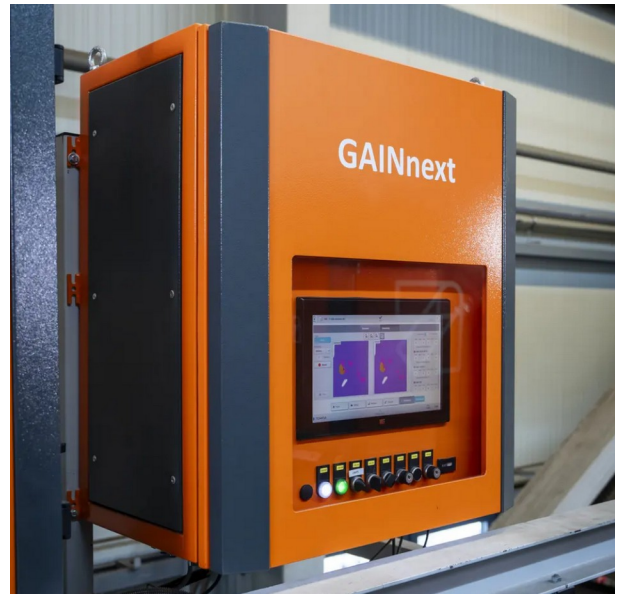
Modern AI refers to the latest developments in the field of deep learning. This is a subcategory of **machine learning** that has found its way into the recycling industry thanks to the advancement of computing power over the last decade.

i. Improving waste sorting accuracy and speed

By using sensors and cameras along with AI technology, recycling facilities can **better identify different types of materials** such as plastics, metals, or paper. This ensures that the sorting process is more precise and faster than traditional methods.

Many recycling companies are striving for a **genuine circular economy**. However, this requires consistently high-quality recovered materials and as many materials as possible to be fed back into the cycle. This is not yet possible with today's processes, and many recovered materials are **downcycled**. To avoid downcycling in favor of using high-quality recyclates, sorting must become more granular. AI will become a catalyst for material circularity. Such advanced technologies significantly improve the sorting and classification of recyclable materials and help to **meet the increasing demand for more recycled content**.

Deep learning is a subcategory of AI. It imitates the way the human brain processes information. It is a special technique within machine learning that uses artificial neural networks that are trained by huge amounts of data, so they recognize and store certain patterns and later apply them to new data. This enables deep learning to solve some of the most complex sorting tasks when **combined with existing sensors**. Instead of replacing hardware components or even machines, modern deep learning technologies can be **retrofitted** with regular software updates as soon as they have been trained by experts. This allows technology suppliers to respond more quickly to customer needs.



Sorting itself will improve considerably. Conventional sorting systems have already achieved a remarkable degree of accuracy and effectiveness, such as advancements in separating single- and multi-layer PET trays. Combining existing optical sorting systems, which are based on near infrared (NIR) and/or visual information sensors (VIS), with deep learning technologies, like with **TOMRA's GAINnext™**, makes granularity sorting more efficient. You can sort by material type

and color and now, thanks to deep learning, also by shape, size, dimensions or other details. It solves previously impossible tasks, such as sorting food-grade vs. non-food-grade PET, PP and HDPE.

ii Optimizing recycling plant operations

Recycling plants now use **AI algorithms** to make their operations more efficient. These algorithms help in managing the flow of materials, optimizing energy usage, and reducing waste. For example, AI can analyze data to predict when machines might need maintenance, **preventing unexpected breakdowns**. AI-powered sorting systems generate huge amounts of data on material composition, sorting efficiency and plant performance. By analyzing this data, operators can identify **optimization opportunities** and act accordingly. Cameras based on deep learning can be placed at key points in the sorting circuit to keep an eye on the entire process and material flow. This allows plant operators to continuously monitor the **quality** of sorted streams, **material loss** and even ensure **compliance** with food recycling regulations.

iii. Enhancing demand forecasting

Another significant application of AI in recycling is through **predictive analytics**, which helps forecast demand for recycled materials. By analyzing historical data and current market trends, AI systems can provide valuable insights into how much recycled material will be needed in the future. This allows recycling plants to adjust their production levels accordingly and **avoid overproduction or shortages**.

E. Manufacturers and suppliers to take note of



VAN DYK Recycling Solutions is a turnkey systems supplier who works with facility owners and operators to offer innovative and profitable recycling solutions. In addition to optical sorters for plastic recycling sorting, they can supply a complete turnkey plastics sorting system including grinders, screens, conveyors, and balers to meet your plastics recycling needs.

TOMRA Recycling provides solutions that enable the circular economy with advanced collection and sorting systems, and food processing by employing sensor-based sorting and grading technology. TOMRA Recycling has pioneered the use of sensor-based technology for automating the recovery and recycling of valuable resources. The flexible sorting systems perform an extensive range of sorting tasks and can sort various types of materials and minerals with the purpose of preparation for recycling, and material upgrading. They have implemented AI with their GAINnext™ AI-powered sorting system.



AES is a leading full-service provider of integrated conveyance systems specializing in scrap, trim, and dust that designs, builds, and maintains economical solutions for more efficient byproduct handling. As a single source for conveyance needs, AES engineers custom scrap systems and works as a partner in process improvement to help businesses achieve greater profitability, productivity, and sustainability in their manufacturing operations.

Jordan Reduction Solutions manufactures rugged industrial shredders, grinders, and granulators exclusively in the U.S.A. They manufacture and stock a full line of OEM replacement parts in their Birmingham, AL facility. All custom mechanical parts for these machines are produced by them in their CNC machine shop and custom fabricating facility. A full test lab is available for demonstration and testing of size reduction, product destruction, and various recycling processes and on site training and installation support is always available.



NICETY Machinery Equipment Co., Ltd. is located in Zibo City, Shandong Province. Founded in 2005, engaged in the development, manufacture and sales of equipment by professional plastic machinery manufacturers. Their plastic machinery mainly includes: mass production extrusion pelletizing,

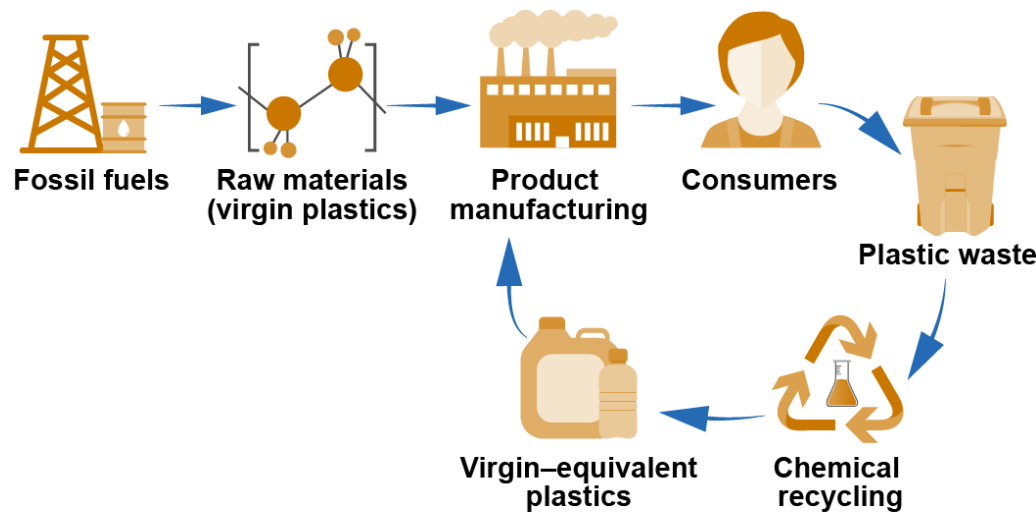
injection molding supporting production lines, vertical blender (500KG-15 tons), horizontal blender (50KG-3 tons), dehydrator, pelletizer, all kinds of vibration screening machine, color powder beating machine, water ring wind ring die head pelletizer, automatic screw feeder, pulling bar suction machine, storage hopper, water tank, crusher, vertical and horizontal drying mixer and other series of products.

EXPO21XX is an online exhibition platform that reaches an extensive worldwide audience of buyers, engineers and business professionals and therefore offers unique **b2b marketing prospects**. EXPO21XX offers an extensive business network connecting the leading technological companies from industrial manufacturing, services, consumer goods and research.



F. Chemical Recycling

Chemical recycling is an **alternative or complementary technology** to current mechanical recycling processes (like sorting and shredding). Chemical recycling technologies use **heat, chemical reactions, or both**, to recycle used plastic into new plastic, fuel, or other chemicals—creating a **closed loop** where plastics can be infinitely recycled to help reduce reliance on landfills.



How does it work?: Chemical recycling can promote a **closed-loop system, known as a circular economy**, wherein plastics are reused rather than discarded in landfills or incinerated. There are three general categories of chemical recycling technologies: **conversion, decomposition, and purification**.

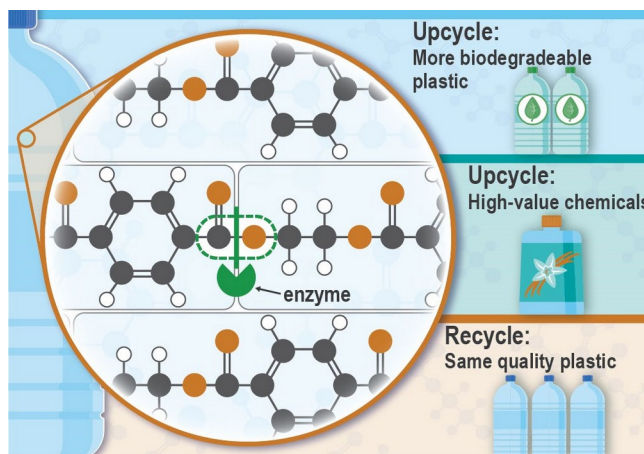
- **Conversion** focuses on converting polymers—long-chain hydrocarbon molecules built from smaller repeating units called monomers—in mixed or sorted plastics into smaller molecules. This can occur through a variety of techniques, including **pyrolysis and gasification**.
 - **Pyrolysis**, sometimes called "plastics to fuel," turns plastic waste into a synthetic crude oil that can be refined into **diesel fuel, gasoline, heating oil, or waxes**. This process involves heating the plastic waste to high temperatures (300-900°C) in the absence of oxygen. Different forms of pyrolysis use different temperatures, pressures, and processing times.
 - **Gasification** also heats plastic waste to high temperatures (500- 1300°C) in a low-oxygen environment to convert plastic waste to **synthesis gas, or "syngas"**. Syngas—a fuel mixture containing mainly hydrogen and carbon monoxide—can be combusted for electric power generation or converted into other fuels or chemicals, such as ethanol and methanol.
- **Decomposition** breaks down polymers in sorted plastics into monomers to produce new plastics. This decomposition can be done with heat or chemicals. Chemical decomposition uses solvents to break the polymers into monomers. Some decomposition technologies use enzymes to break down polymers at temperatures as low as room temperature, resulting in less energy consumption.
- **Purification** uses solvents to separate polymers from additives or contaminants. Unlike other types of chemical recycling, purification does not break or modify the polymer. Purification may be used with mixed or sorted plastics.

It should be noted that adopting chemical recycling technologies faces hurdles, like **high startup costs** and few incentives for investing in innovation.

G. Biorecycling

Biological recycling, or biorecycling, is an emerging technology that uses microbes, such as bacteria or fungi, to **break down plastic into its basic components** for reuse. In contrast, the most commonly used recycling technology—mechanical—grinds, washes, and re-granulates the plastic. Chemical recycling, another emerging technology, uses heat, chemical reactions, or both to recycle plastic waste. Industry is considering advanced technologies such as biorecycling and chemical recycling as complements or alternatives to mechanical recycling. With mechanical recycling, **repeated processing eventually results in lower quality** plastic that is discarded in landfills.

How does it work?: The microbes' specialized proteins called **enzymes** degrade plastic into its most basic chemical components, or **monomers**. The monomers can then be recombined into plastics of the same type and quality. In a process called upcycling, the monomers are recombined into different materials or chemicals with more desirable qualities, such as plastics that are more biodegradable than originally made or high-value chemicals, such as vanillin found in vanilla flavorings.



While some microbes have naturally evolved enzymes that can degrade certain plastics, the process **in nature can take weeks** or more. To make biorecycling viable at industrial scale, scientists can alter, or engineer, the natural enzyme to degrade plastics faster, reducing the time needed for complete degradation to **hours**.

Entities that hope to engage in biorecycling could face a **complicated legal landscape** that may pose a challenge for the emerging technology. At the federal level, depending on the specifics of the process, aspects of biorecycling or the wastes that may result from that process might be **governed by various laws as amended**. In addition, states, tribal organizations, municipalities, and other stakeholders—such as nonprofit organizations, businesses, and other entities—can also play important roles in **regulating or supporting recycling** in the U.S.

Section III. Add-Ons

NICETY has a blog with helpful articles, resources, and ideas for more in-depth upscaling solutions and professional support: <https://nicetymachine.com/blog/>

A. Recycling more than just bottles

Conduct market research to determine who your competitors are, how successful they are, and whether your area has enough volume to support recycling other materials, such as **aluminum, glass, steel, paper, computers, or batteries**. Are there any materials that the current recycling companies are ignoring? Make sure your research assists you in selecting both profitable materials and a competitive niche.

B. Customer interaction and retention

i. Customer data management (CDM)

CDM is the way businesses keep track of their customer information and survey their customer base in order to obtain feedback. CDM includes a range of **software or cloud computing applications** designed to give large organizations rapid and efficient access to customer data. Surveys and data can be centrally located and widely accessible within a company, as opposed to being warehoused in separate departments. CDM encompasses the **collection, analysis, organizing, reporting and sharing of customer information** throughout an organization. Businesses need a thorough understanding of their customers' needs if they are to **retain and increase their customer base**. Efficient CDM solutions provide companies with the ability to deal instantly with customer issues and obtain immediate feedback. As a result, customer retention and customer satisfaction can show marked improvement.

- should provide a cost-effective, user-friendly solution for marketing, research, sales, human resources and IT departments
- enables companies to create and email online surveys, reports and newsletters
- encompasses and simplifies customer relationship management (CRM) and customer feedback management (CFM)

ii. Social media marketing

In addition to marketing your recycling business using traditional methods, such as **posting fliers, advertising in print, creating a website and posting in online news publications**, consider digital and mobile marketing. Social media advertising, for example, is often more cost-effective than traditional marketing tactics and reaches a **far broader audience**, since most U.S. adults regularly use social media and check their accounts multiple times a day, enabling you to connect to an otherwise difficult-to-reach customer base.

What are the benefits of opening social media accounts?

- Increases your brand awareness.
- Helps you build a relationship with your potential customers.
- Drives traffic to your website.
- Generate potential leads for you.

You can create social media accounts for free. Then, come up with a social media strategy that outlines your **goals, target audience, and the platforms** you will use. Later, add engaging content, such as **infographics, photos, and videos** that highlight the importance of waste management and showcase your services. As you continue, engage politely with your audience by **responding to comments**, messages, and posts.

iii. Mobile apps for tracking recycling progress

In today's digital age, mobile apps have become an integral part of our lives. Recognizing this trend, some manufacturers have incorporated **mobile app functionality into their automatic plastic bottle recycling machines**. These apps provide users with a convenient way to track their recycling progress and **engage with the recycling process**.

By downloading the app, individuals can easily monitor the number of bottles they have recycled, track their environmental impact, and set personal goals for recycling. This **gamification** aspect encourages users to continue recycling and stay engaged in sustainable practices.

iv. Increasing public awareness about the benefits

One of the challenges in implementing automatic plastic bottle recycling machines is the need to increase public awareness about their benefits. Many individuals may not fully understand how these machines contribute to environmental sustainability or how they can actively participate in the process.

Educational campaigns should be launched to raise awareness about the importance of recycling plastic bottles and how automatic recycling machines play a significant role in this endeavor. These campaigns can employ various strategies such as **social media outreach, community workshops, informative videos, or even partnering with local schools** to educate children who can then influence their families' behavior.

Highlighting the positive impact of recycling on reducing pollution, conserving resources, and mitigating climate change can help garner public support for installing more of these machines. Emphasizing the convenience factor – making it clear that using these machines is easy and hassle-free – can encourage individuals to actively engage in recycling efforts.

v. Search engine optimization (SEO)

By optimizing your website, you can improve your visibility to potential customers who are searching for recycling services in your area. This ultimately generates more leads and sales.

How to implement it:

- Firstly, start by conducting keyword research to determine the terms your potential customers are searching for.
- Optimize your website by including these keywords in your page titles, meta descriptions, and content.
- Ensure that your website is mobile-friendly, fast-loading, and easy to navigate, as these factors can impact your SEO rankings.

Section IV. Further resources

A. Books

The Revenue Growth Habit: The Simple Art of Growing Your Business by 15% in 15 Minutes Per Day
by Alex Goldfayn

The 10X Rule: The Only Difference Between Success and Failure by Grant Cardone
Bottle Recycling Service Business Plan: Deluxe Edition 2023 by Scott Proctor ebook
The E-myth revisited why most small businesses don't work and what to do about it by Michael E. Gerber
How to grow your small business: a 6-step plan to help your business take off by Donald Miller
Deduct It!: lower your small business taxes by Stephen Fishman
HBR's 10 must reads on AI, analytics, and the new Machine Age
Small Business by Eric Tyson

B. Contacts

Metro New York District Offices: New York City—Main office
26 Federal Place, Suite 3100, New York, NY 10278 Phone: 212.264.4354
Services we provide Our office provides help with SBA services including funding programs, counseling, federal contracting certifications, and disaster recovery. We can also connect you to our partner organizations, lenders, and other community groups that help small businesses succeed. Hours of operation: Monday through Friday, from 8:00 a.m. to 4:30 p.m. Closed on all federal holidays. Serving: Kings, Queens, Bronx, Dutchess, New York, Orange, Putnam, Rockland, Richmond, Sullivan, Ulster, and Westchester counties.
NYC Department of Small Business Services—NYC Business Solutions
Telephone: 311 https://www1.nyc.gov/site/sbs/index.page
The NYC Business Solutions centers offer a suite of services to help one start, operate, and expand a business in New York City. Regardless of the size or stage of the business, NYC Business Solutions can help individuals access resources that meet their needs as their

business continues to grow and expand. NYC Business Solutions can help one develop a business plan, access business courses, find financing, hire workers, train employees, navigate government regulations, access government incentives programs, and obtain minority and women-owned business enterprise certification for free.

For more information about starting a business, visit the NYC Business Solutions website listed above to download helpful guides and checklists or speak with a transitional services coordinator to obtain copies of these resources. A detailed assistance request form can also be submitted online at <https://www1.nyc.gov/site/sbs/businesses/contact.page>.

Bronx Center

400 East Fordham Road, 7th Floor, Bronx, NY 10458

Brooklyn Center

9 Bond Street, 5th Floor, Brooklyn, NY 11201

Queens Center

90-27 Sutphin Blvd., 4th Floor, Jamaica, NY 11435

Lower Manhattan Center

14 Wall Street, 17th Floor, New York, NY 10005

Upper Manhattan Center

361 West 125th Street, 2nd Floor, New York, NY 10027

Staten Island Center

120 Stuyvesant Place, 3rd Floor, New York, NY 10301

Washington Heights Center

560 West 181st Street, 2nd Floor, New York, NY 10033

U.S. Small Business Association

26 Federal Pl., Suite 3100 New York, NY 10278

Phone: 212-264-4354

The United States Small Business Administration (SBA) is an independent agency of the United States government that provides support to entrepreneurs and small businesses. The agency's activities have been summarized as the "3 Cs" of capital, contracts and counseling. [

SCORE New York City

26 Federal Plaza, Room 3100, New York, NY 10278

Telephone: 212.264.4507

www.score.org/newyorkcity

Provides free business counseling and advice to all types of businesses in all stages of development, from idea to start-up to success, from a national network of retired and working volunteers. Offers low-cost online workshops. Also offers a veterans entrepreneurship program. Visit the website to search for a counselor or to request a free mentoring appointment.

Empire State Development—Division of Minority and Women's Business Development

Telephone: 646.846.7364

<https://esd.ny.gov/doing-business-ny/mwbe>

The Division of Minority and Women's Business Development promotes the participation of minority and women-owned businesses in the New York City procurement process.